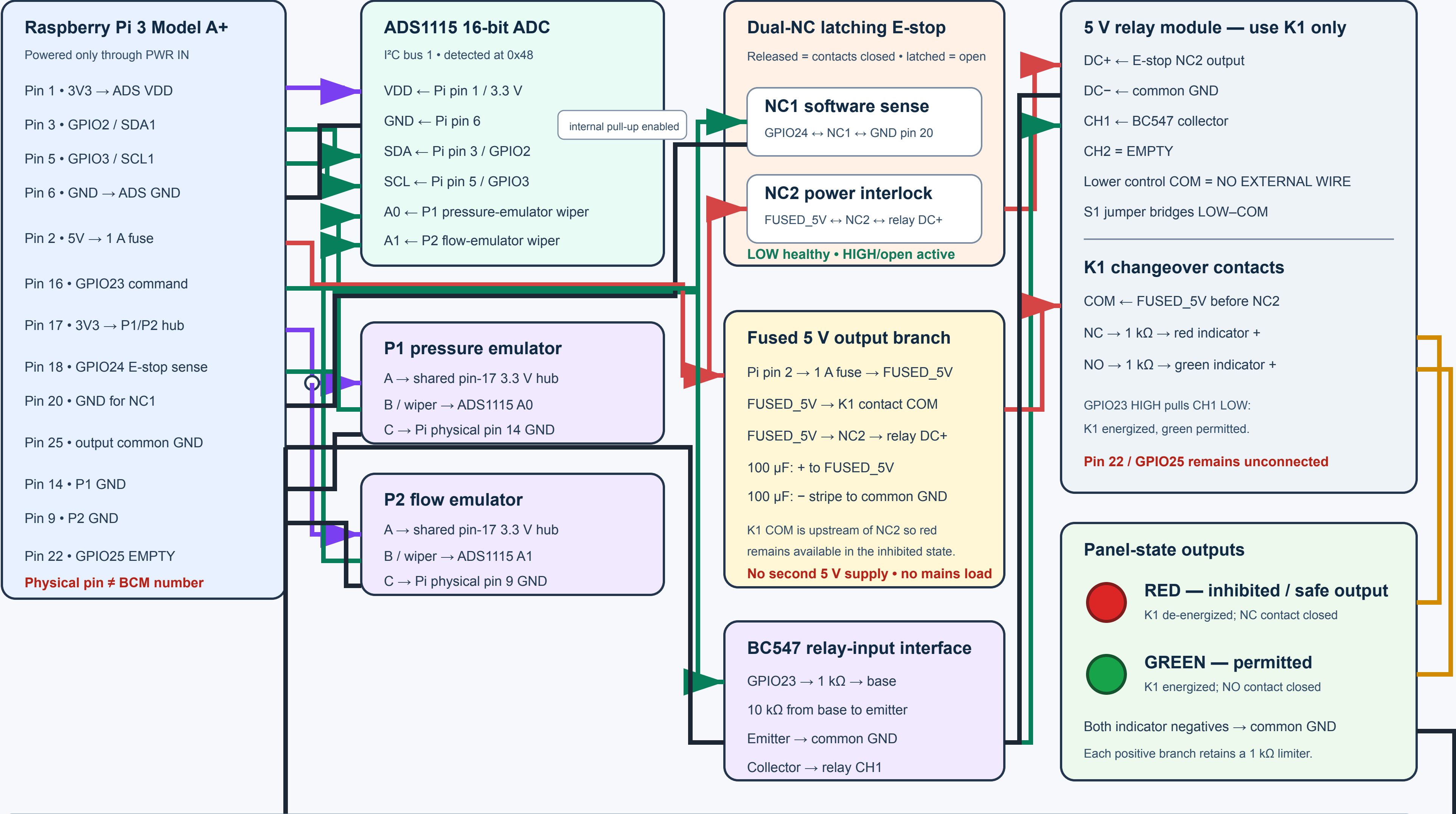


CertaRig Phase 3 — final as-built dry-bench circuit

Verified 12 September 2026 • low voltage only • Raspberry Pi 3 A+ • relay channel K1 • no hydraulic or mains load



Verified operating truth table			
E-stop released, before reset:	board powered • K1 OFF • RED ON • GREEN OFF	E-stop latched / open:	relay DC+ removed • GPIO23 LOW • K1 OFF
Reset + permit accepted:	GPIO23 HIGH • K1 ON • RED OFF • GREEN ON	Release after trip:	K1 remains OFF until explicit reset + new permit

Submission boundary: this diagram represents the tested low-voltage dry bench only. Hydraulic commissioning and real actuator loads remain capstone work.